Nhill 2.0 — Configuration guide

Français : ../fr/guide-configuration.md

Three levels of settings, from infrastructure to daily use:

1. **Server** — file `/etc/nhill/server.env` (§ 1);
2. **Applications** — build options and environment variables (§ 2);
3. **Platform and businesses** — Nhill Console and Nhill Business screens (§ 3).

The marketing website has its own, separate configuration: `website/README.en.md` § 3.

1. Server: `/etc/nhill/server.env`

Secret-free template: `config/server.env.example` (installed in `/usr/share/nhill/examples/`). The file belongs to `nhill`, mode **600**: the server refuses to start if others can read it (`NH_ALLOW_INSECURE_CONFIG` lifts this check, for tests only). Every key can also be given as an environment variable. After a change: `sudo systemctl restart nhill-server`, then `sudo -u nhill nhill-server --config /etc/nhill/server.env check`.

Key	Default	Purpose
`NH_DATABASE_URL`	—	connection of the application role `nhill_app` (subject to RLS). Written by the setup wizard
`NH_MIGRATION_DATABASE_URL`	—	connection of the owner role `nhill_owner` (migrations at start-up)
`NH_DB_PROVIDER`	—	`supabase` or `postgresql`, written by setup
`NH_DB_POOL_SIZE`	= `NH_THREADS`	open connections to the database
`NH_LISTEN_ADDRESS`	`127.0.0.1`	listen address (stay on loopback behind Caddy)
`NH_LISTEN_PORT`	`8080`	listen port
`NH_THREADS`	`16`	HTTP worker threads
`NH_TRUST_PROXY`	`0`	`1` behind a trusted proxy: the client address is the **rightmost** `X-Forwarded-For` entry (2.0)
`NH_PUBLIC_BASE_URL`	—	public address `https://api.<domain>` (webhooks, payment return pages)
`NH_SECRET_KEY`	—	64 hexadecimal characters (`openssl rand -hex 32`), generated by the package. **Back it up separately**
`NH_WEBHOOK_SECRET_<PROVIDER>`	—	webhook secret (≥ 32 characters) of a generic provider
`NH_ACCESS_TOKEN_MINUTES`	`15`	access token lifetime
`NH_REFRESH_TOKEN_DAYS`	`30`	refresh token lifetime (rotation, reuse detection)

Key	Default	Purpose
`NH_ARGON2_OPS`	`3`	Argon2id cost (iterations)
`NH_ARGON2_MEM_KIB`	`65536`	Argon2id memory (64 MiB)
`NH_JOBS_INTERVAL_SECONDS`	`300`	background jobs (code rotation, subscriptions); `0` disables them
`NH_EMAIL_TRANSPORT`	`log`	only transport available: e-mails are written to the log, **nothing is sent**
`NH_PROFILE_DIR`, `NH_I18N_DIR`, `NH_FONT_DIR`, `NH_MIGRATIONS_DIR`	`/usr/share/nhill/…`	data files installed by the package
`NH_SETUP_DB_PASSWORD`	—	only for command-line `nhill-server setup`; never kept

Recommendations: do not lower `NH_ARGON2_*` below the defaults; measure sign-in time after raising them. Keep `NH_LISTEN_ADDRESS=127.0.0.1` and expose only Caddy. `NH_TRUST_PROXY=1` with no proxy in front would let a client choose the address seen by rate limits.

Supabase

Use the **Session pooler** (5432). No `anon` or `service_role` key is used. The Supabase security advisor can be checked after setup; Nhill creates a `nhill` schema not exposed by the REST API.

2. Applications

2.1 Build options (CMake)

Option	Default	Effect
`NHILL_SERVER_URL`	`http://127.0.0.1:8080`	server address compiled into the applications — **HTTPS in production**
`NHILL_BUILD_SERVER`	`ON` on Linux	server (Linux only)
`NHILL_BUILD_CLIENT` / `NHILL_BUILD_TESTS`	`ON`	applications / tests
`NHILL_WARNINGS_AS_ERRORS`	`OFF`	warnings as errors (CI)

The build scripts accept `--server-url https://api.<domain>`.

2.2 Environment variables (desktop)

Variable	Values	Effect
`NHILL_SERVER`	URL	server to use instead of the compiled address (trials, staging)
`NHILL_LANG`	`fr`, `en`	interface language at start-up
`NHILL_THEME`	`light`, `dark`, `system`	theme at start-up
`QT_SCALE_FACTOR`	e.g. `1.25`	global enlargement (standard Qt variable)

The `NHILL_TOUR_*` and `NHILL_SETUP_TOUR` variables are only used for automatic screenshots (development).

The client refuses `http://` addresses except `localhost`, `127.0.0.1`, `::1` and `10.0.2.2` (Android emulator).

2.3 User preferences

Settings › My account: language, theme. Window size and position are computed at every start from the available screen.

3. Platform and businesses

3.1 Nhill Console (creators)

Screen	Settings
Platform settings	trial length (30 days by default), grace period, reminder, establishment code rotation (30 days), limits, payment instructions
Plans and prices	plans, limits (users, POS tills, warehouses), features, **prices per duration and currency**
Promo codes	launch or partnership discounts
Payment providers	credentials (encrypted, never shown again), test/production mode, webhook address

3.2 Nhill Business (business)

Tab	Main settings
Business	identity, legal identifiers, logo, document language, payment terms, tax rounding, footer, negative stock
Taxes and accounts	tax rates (16 % VAT in the DRC profile; a rate cannot be edited), cash and bank accounts, dated exchange rates
Point of sale	tills, authorised cashiers, payment methods, maximum discount without approval, 58/80 mm receipt
Team	roles and invitations

3.3 Country profiles and templates

`server/profiles/*.json`: tax profiles (rates, mentions, numbering) synchronised into the database at start-up. An issued document keeps the version of the profile applied. Details: `docs/modeles-et-profils.md` (French). The tax values of the profiles must be checked by a professional before use (see legal-parameters.md).

4. Password rule (2.0)

Item	Value	Where
Minimum length	**8 characters** (Unicode: an “é” counts as one)	`common/include/nhill/validation.hpp` → `kMinPasswordLength`
Maximum length	128 bytes	`kMaxPasswordBytes`
Refused	common passwords (built-in list), same as the e-mail, repetitive	`common/src/validation.cpp`
Periodic expiry	**none**	—
Storage	Argon2id (libsodium), never in plain text	server
Two-factor authentication	mandatory for Nhill Console; not imposed on businesses	—

The same function is called by the applications (before sending) and by the server (which decides). Changing the rule requires rebuilding the server **and** the applications.